Towards A Universally Causal Reasoner

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Abstract

As large language models (LLMs) are increasingly deployed in high-stakes domains, we aim to advance their capabilities in causal reasoning so that they can naturally discover and utilize causal structures when addressing real-world challenges. Existing efforts fail to produce such universally causal reasoners due to two fundamental bottlenecks: a lack of comprehensive causal training data, and an insufficient understanding of training algorithms that induce generalizable causal skills. We propose a three-step approach to bridge this gap: (1) a unified training data framework spanning diverse causal levels and task domains; (2) an exploration of online, on-policy, and verifier-based algorithms to identify the most supervision-efficient training methods; and (3) a rigorous evaluation of the causal reasoner's "universality" in flexible real-world applications and general reasoning tasks, coupled with a generalization analysis of how causality-centered training systematically reshapes external reasoning patterns and internal representations.

1 Introduction

This proposal aims to build an LLM agent that can **universally** understand and reason about the world in a causal way, so that it can be flexibly deployed in various high-stakes domains, such as medicine and law, with consistently ensured reliability and interpretability [Kiciman et al., 2024]. Specifically, we define the universality of a causal reasoner along two axis: (1) **within formal causality**: the LLM must master a comprehensive scope of formal causal theories and their corresponding reasoning skills across the three levels of Causal Hierarchy [Pearl, 2009], which can be exemplified by excellent performance on dedicated causal reasoning benchmarks (e.g., CLadder Jin et al. [2023] and Counterbench Chen et al. [2025]). (2) **beyond formal causality**: the reasoner should autonomously and flexibly adopt a causal mindset when dealing with real-world challenges where causality is not explicitly defined but implicitly fundamental (e.g., in clinical diagnosis or scientific discovery), so as to discover and utilize their hidden causal structures.

However, existing research has failed to produce such universally causal reasoners due to two fundamental bottlenecks. First, there is a critical lack of **training data**. Current efforts focus on benchmarking LLMs on a specific aspect of causal reasoning, which either produces no training datasets [Chen et al., 2025, Jin et al., 2023, 2024], or only constructs synthetic data in a narrow scope with limited diversity or scalability [Dong et al., 2025, Vashishtha et al., 2025]. Second, it is still unclear what **training algorithms** can instill generalizable causal skills in the most data-efficient way. Traditional supervised fine-tuning (SFT) remains the main focus of existing efforts [Dong et al., 2025], leaving online [Lambert et al., 2025], on-policy [Lu and Lab, 2025], and test-time [Zhang et al., 2025] algorithms that are inherently more supervision-efficient largely unexplored. In this proposal, we aim to address these challenges, developing a universally causal reasoner with causality serving as a generalizable meta-reasoning capability.

Research Objectives. We propose a three-step approach (Figure 1):

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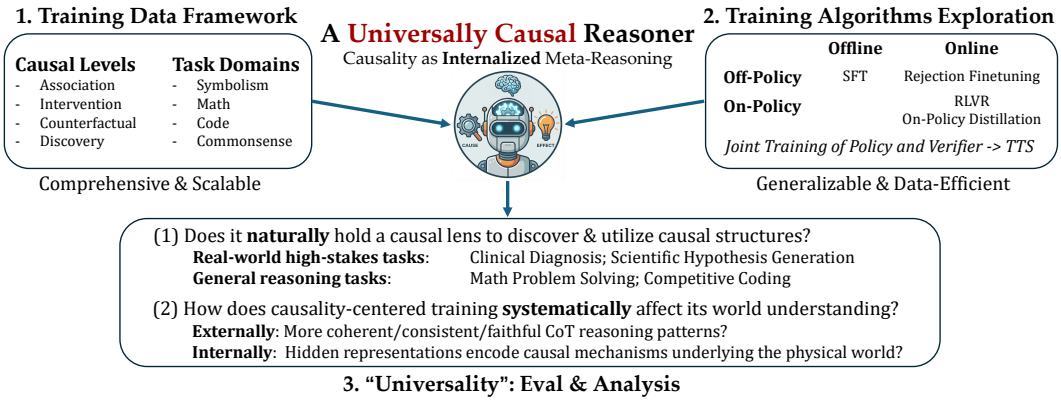

Figure 1: Overview of our proposed approach.

1. **Data Framework**: We will develop a unified data generation framework covering two dimensions–causal levels and task domains–towards a suite of diverse training data with fine-grained difficulty control. We will also conduct controlled experiments to probe the transfers and trade-offs between both dimensions, thus guiding the composition of the final dataset.
2. **Algorithmic Exploration**: Based on the dataset from the previous step, we will start with SFT, and move forward to explore a variety of online and on-policy training algorithms for instilling generalizable causal reasoning skills. We will also explore joint training of reasoners and verifiers for test-time scaling.
3. **Generalization Analysis**: We will eventually assess the "universality" of our causal reasoners, especially in (i) real-world high-stakes tasks that require flexible applications of causal reasoning, such as clinical diagnosis and scientific hypothesis generation, and (ii) general reasoning tasks that are not explicitly causal, such as math problem solving and competitive coding. We will analyze both external reasoning patterns and internal model representations to probe the effect of causality-centered training on our reasoners' understanding about the world.

Expected Impact. Success in this project will produce a reasoner that internalizes causality as a meta-reasoning capability, and naturally holds a causal lens to understand and address real-world challenges. It can promote trustworthiness and explainability in various real-world high-stakes domains, supporting reliable clinical diagnosis, truthful legal judgments, prudent policy making, and grounded scientific discovery. Moreover, it will provide insights into how causality-centered training systematically affects a language agent's internal reasoning representations that model the causal mechanisms underlying the physical world. This will provide a valuable interface to understand and improve language agents for world modeling and real-world scientific discovery.

2 Proposed Approach

2.1 A Unified Two-Dimensional Training Data Framework

A universally causal reasoner needs to master a comprehensive distribution of causal reasoning skills. We will approach this comprehensiveness by defining a two-dimensional taxonomy that consists of four **causal levels** (association, intervention, counterfactual, and causal discovery) and three **task domains** (symbolism, math, and code). We will then develop a unified training data generation framework across both dimensions, based on combinations of *causal graphs*, *causal query types*, and *task domains*. Specifically, this framework will use a directed acyclic graph (DAG) sampler [Lampinen et al., 2023] to obtain causal graphs with diverse structures and fine-grained complexity control. Each causal graph will be automatically converted into a structured causal model (SCM) by introducing linear dependencies and noise variables. Next, each SCM will be paired with query types specific to different causal levels, resulting in graph-based reasoning questions in the symbolic domain. Finally, they will be translated into math and code domains by assigning numerical

contexts to graph components, or treating causal graphs as data flow graphs (DFG) and control flow graphs (CFG) that encode functional semantics.

Apart from comprehensiveness, it is also crucial to consider the transfers and conflicts across subgroups of data, which have been revealed among different types of reasoning [Dai et al., 2025, Huan et al., 2025], and also substantiated by our initial experiments in the context of causality (Figure 2). Therefore, our data framework will conduct controlled experiments to probe the trade-offs within and between both dimensions, in order to identify the causal levels and task domains that generalize best. These exploratory experiments will provide valuable guidance on tuning the compositions of different subgroups in the final training data mixture.

2.2 Exploring Supervision-Efficient Training Algorithms

With a curated dataset, we will then investigate the most data-efficient training algorithms for instilling generalizable causal reasoning. While SFT remains the dominant approach in existing causal LLM research, its offline and off-policy nature often leads to limited generalization despite high annotation costs. Therefore, we will pivot towards online and on-policy algorithms, especially reinforcement learning from verifiable rewards (RLVR) [Lambert et al., 2025, Shao et al., 2024]. Our data framework is uniquely positioned to support this, as it inherently provides verifiable ground-truth signals for reward calculation. Furthermore, to mitigate the rollout overhead and sparse reward challenges of standard RL, we will explore On-Policy Distillation [Lu and Lab, 2025]. By allowing the student agent to sample its own reasoning trajectories while receiving token-level feedback from a stronger teacher, this method effectively combines the exploration benefits of RL with the supervision density of SFT.

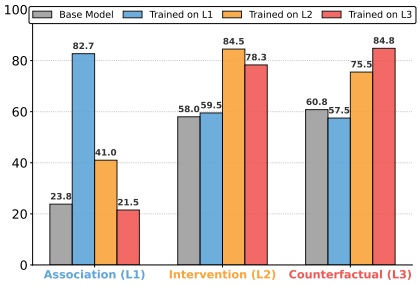

Figure 2: Initial results demonstrating transfers and conflicts across causal levels in the CLadder dataset [Jin et al., 2023]. Training on counterfactual (level 3) data leads to significant performance gains in intervention (level 2) tasks, but meanwhile losses in association (level 1) tasks (and vice versa), suggesting positive transfers between level 2 and 3, and negative conflicts between level 1 and 3.

Beyond training the causal reasoner in isolation, we will also investigate the joint training of reasoners and external verifiers. This approach creates a synergistic mechanism where a verifier steers the reasoning process, identifying and correcting flawed causal logic during test-time scaling. By co-evolving the reasoner and the verifier iteratively, we aim to achieve a higher degree of robustness in reasoning, ensuring the reasoner can adhere to the causal mindset even when faced with complex high-stakes challenges.

2.3 "Universality" Evaluation and Generalization Analysis

To rigorously assess the "universality" of our reasoner, we will expand our evaluation beyond standard causal benchmarks to two frontiers of generalization. The first contains real-world high-stakes tasks that demand the flexible application of causal reasoning, such as clinical diagnosis, legal judgment, policy making, and scientific hypothesis generation [Liu et al., 2025]. The second targets general reasoning tasks that are not explicitly causal, including complex math problem solving and competitive coding. Our core hypothesis is that a universally causal reasoner will internalize causality as a meta-reasoning capability, and naturally hold a causal lens to discover and utilize causal structures when addressing real-world challenges.

Beyond performance numbers, we will carry out analyses on two levels to understand the mechanisms of this generalization. Externally, we will examine the reasoner's Chain-of-Thought (CoT) reasoning patterns, evaluating whether causality-centered training systematically leads to more consistent and faithful reasoning trajectories. Internally, we will probe the reasoner's hidden representations to determine how this training systematically reshapes its perception and modeling of the physical world. We will provide critical insights into how language agents internalize the causal mechanisms underlying reality, serving as a valuable interface for understanding and improving world modeling and real-world scientific discovery.

3 Timeline

Weeks 1–2: Finalize the dataset construction pipeline. Building on the past three months of development work, we will finalize our unified two-dimensional data generation framework, ensuring comprehensive coverage across all four causal levels and three task domains with fine-grained difficulty control. We will also complete the controlled experiments probing transfers and conflicts across data subgroups to guide the composition of the final training data mixture.

Week 3: Implement GRPO-based RLVR training pipeline. We will implement a Group Relative Policy Optimization (GRPO) [Shao et al., 2024] based reinforcement learning from verifiable rewards (RLVR) pipeline to train a universally causal reasoner from the constructed data, leveraging the verifiable ground-truth signals inherent in our data framework for reward calculation.

Weeks 4–6: Generalization analysis and paper writing. We will conduct the universality evaluation across real-world high-stakes tasks and general reasoning benchmarks, perform generalization analysis on both external reasoning patterns and internal representations, and write the final paper.

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